

LAB : 3

Title : decision control structures in cpp

1. Objective

The purpose of this lab is to understand and implement:

- Comparison/Relational Operators
- Logical Operators
- Decision-making structures (if, if-else, else-if)

2. Theory

2.1 Relational Operators

Relational operators compare two values and return a boolean result (true or false).

Examples:

- < : Less than
- > : Greater than
- <= : Less than or equal to
- >= : Greater than or equal to
- == : Equal to
- != : Not equal to

2.2 Logical Operators

Logical operators combine or negate conditions.

- ! : NOT
- && : AND
- || : OR

2.3 Decision Control Structures

- **if statement:** Executes a block if condition is true.
- **if-else statement:** Executes one block if condition is true, otherwise another block.
- **else-if statement:** Allows multiple conditions to be checked sequentially.

3. Programs and Results

Task 1: Comparison Operators

```
#include <iostream> //tells the compiler to include input output stream library

using namespace std; //tells the compiler to use standard namespace std

int main () { // main function starting point

    int a, b; // variables declaration

    cout << "Enter two numbers: "; // output prompt for user

    cin >> a >> b; // takes input from user

    if (a == b) // if condition provided

        cout << "Both numbers are equal." << endl; // output message to display

    else if (a > b) // 2nd condition

        cout << "First number is greater." << endl; // message sent to display 2

    else // else condition

        cout << "First number is smaller." << endl; // output message to display

    return 0; } // tells the computer that program has ended
```

Sample Output:

Enter two numbers: 10 20

First number is smaller.

Task 2: Logical Operators

```
#include <iostream>    //tells the compiler to include input output stream library
using namespace std;   //tells the compiler to use standard namespace std

int main() {           // main function starting point
    int num;           // variable declaration
    cout << "Enter a number: "; // output prompt for user
    cin >> num;         // takes input from user

    if (num > 0 && num < 100) // if condition provided within a range
        cout << "Number is between 0 and 100." << endl; // output message
    else // else condition
        cout << "Number does not meet the condition." << endl; // output message

    return 0;          // tells the computer that program has ended
}
```

Sample Output:

Enter a number: 50

Number is between 0 and 100.

Task 3: If Statement

```
#include <iostream>    //tells the compiler to include input output stream library

using namespace std;   //tells the compiler to use standard namespace std


int main() {           // main function starting point

    int num;           // variable declaration

    cout << "Enter a number: "; //output prompt for user

    cin >> num;         // takes input from user


    if (num > 0)        // if condition provided that num will be greater than 0

        cout << "Number is positive." << endl; //output message


    return 0;          // tells the computer that program has ended

}
```

Sample Output:

Enter a number: -5

(no output since condition is false)

Task 4: If-Else Statement

```
#include <iostream>    //tells the compiler to include input output stream library

using namespace std;   //tells the compiler to use standard namespace std


int main() {           // main function starting point

    int age;           // variable declaration

    cout << "Enter your age: "; // output prompt for user
```

```

cin >> age;                                // takes input from user

if (age >= 18)                              // if condition which checks age
    cout << "You are eligible to vote." << endl;    // output message
else                                        // else condition
    cout << "You are not eligible to vote." << endl; //output message

return 0;                                // tells the computer that program has ended
}

```

Sample Output:

Enter your age: 16

You are not eligible to vote.

Task 5: Else-If Statement

```

#include <iostream>    //tells the compiler to include input output stream library
using namespace std; //tells the compiler to use standard namespace std

int main() {          // main function starting point
    int num;          // variable declaration
    cout << "Enter a number: "; // output prompt for user
    cin >> num;        // takes input from user

    if (num > 0)       // if condition which checks num is greater than 0
        cout << "Number is positive." << endl;    // output sent to display
}

```

```
else if (num < 0)                //2nd condition
    cout << "Number is negative." << endl; //2nd message sent to display
else                             //else condition
    cout << "Number is zero." << endl;    // message sent to display

return 0;                        // tells the computer that program has ended
}
```

Sample Output:

Enter a number: 0

Number is zero.

4. Discussion

- Relational operators allow comparison between values, essential for decision-making.
- Logical operators combine conditions, making programs more flexible.
- Control structures (if, if-else, else-if) enable branching logic, allowing programs to respond differently based on input.

5. Conclusion

In this lab, I successfully implemented relational operators, logical operators, and conditional statements in C++. These concepts are fundamental for writing programs that make decisions based on user input.